

ISyE 7406 - Data Mining and Statistical Learning

Georgia Institute of Technology

School of Industrial and Systems Engineering

Evaluating Variable Subsets and Classifiers for Engine Fault Detection

Author: Nicole Athanasiou
Student ID:
Email:

Date: July 24, 2026

1 Introduction

This project aims to classify engine faults using engine and gas-sensor measurements – a topic important to manufacturers, customers, and researchers. Accurate fault classification is crucial for identifying which variables are most influential to engine failure. These insights will highlight which sensor measurements should be prioritized in onboard diagnostic systems, preventing spontaneous combustion or electrical malfunction. Moreover, improved fault detection improves vehicle safety and reduces unanticipated repair costs – a priority for many customers. For researchers, evaluating a range of classification models supports development of more reliable vehicle-health monitoring technologies. The contributions strengthen a broader effort to build smarter, safer, and more efficient automotive systems.

A previous analysis provides baseline results using several machine-learning models, supported by exploratory analysis through correlation heatmaps and boxplots (referred to as the "baseline" throughout this paper) [11]. However, variable selection is not performed despite clear multicollinearity and known physical relationships among variables. Models used include Logistic Regression, Decision Trees, Random Forests, SVC, KNN, Naïve Bayes, and a Feed-Forward Neural Network. The work offers useful baselines but leaves methodological gaps.

This project will improve upon past analysis by using a more rigorous methodology. First, three different sets of variables will be compared – engine measurements, gas-sensor measurements, and variables chosen through feature selection method Ridge Regression. Because engine behavior may drive exhaust-gas characteristics, the separation is essential for determining whether combined modeling from previous analysis is redundant or adds value. The modeling itself will be expanded to include LDA, QDA, and Gradient Boosting, as well as the original 7 models. Moreover, this analysis incorporates cross-validation and statistical significance testing with Wilcoxon to provide reliable model performance evaluation. By evaluating a range of variable combinations and models and thoroughly testing performances, the takeaways will be more reliable for manufacturers, customers, and researchers.

2 Data Sources

The dataset – EngineFaultDB – includes faults for automotive engine characteristics ranging from throttle position to exhaust gas emissions [9]. The data comes from a C14NE spark ignition engine in a laboratory setting and has 55,999 rows and 15 variables. 14 variables are continuous; Fault is discrete with values 0, 1, 2, and 3, where 0 represents no fault. The other fault types are described in Figure 1. The classes are mildly imbalanced; the distribution ranges from 19% to 28% (Figure 2). The predictors are left-skewed in the histograms; normality is not met (Appendix A.1). Multicollinearity is clearly present from the correlation heatmap, especially within engine-only and gas-only variables (Appendix A.2). The VIF table quantifies the levels of multicollinearity with AFR, Lambda, RPM, and Speed being the highest (Appendix A.3). Some of these correlations are clearly explainable; Lambda is calculated directly from AFR. More on correlations is discussed in Appendix A.2. Throughout data collection, outliers deemed inconsistent or erroneous were excluded. Based on the EDA done in the publication, the variables have high variability, implying the dataset captures a range of engine demands, driving conditions, and combustion scenarios [11].

3 Proposed Methodology

The main research question is: How do classification models of varying complexity perform across the three analysis tracks (engine-only, gas-only, combined-with-selection)? From this, past results and the best model in terms of accuracy,

Fault type	Fault name	Conditions
Fault 1	Rich mixture	Incorrect sensor performance High fuel pressure Defective injector Faulty pressure regulator Clogged air filter Clogged fuel return line
Fault 2	Lean mixture	Incorrect sensor performance Low fuel pressure Defective injector Faulty pressure regulator
Fault 3	Low voltage	Worn spark plugs Faulty ignition cables Defective coil Faulty sensor wiring

Figure 1: Fault Definitions [11]

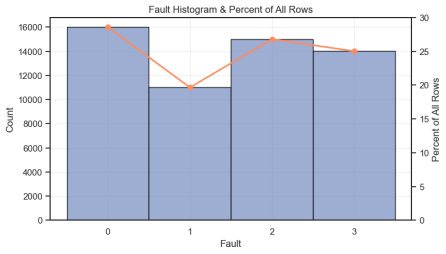


Figure 2: Class Balance: The classes are mildly imbalanced, ranging from 19% to 28%.

interpretability, and real-world applications can be evaluated thoroughly.

3.1 Variable Combinations

3.1.1 Definitions

Variable definitions are below and taken directly from the original paper [11].

- **Manifold Absolute Pressure (MAP):** A measure of the pressure within the intake manifold, crucial for the engine's electronic control system to regulate fuel injection and ignition timing, measured in kilopascals (kPa).
- **Throttle Position Sensor (TPS):** Provides information about the position of the throttle, influencing fuel injection, ignition timing, and other engine parameters. It is measured as a percentage.
- **Force:** Represents the engine's torque or rotational force. It is measured in newtons (N).
- **Power:** Quantifies the rate at which work is done or energy is transferred in the engine. It is measured in kilowatts (kW).
- **Revolutions Per Minute (RPM):** Indicates the engine's speed, detailing how many times the engine's crankshaft rotates per minute.
- **Fuel consumption (l/h):** Illustrates the engine's fuel consumption rate.
- **Fuel consumption (l/100km):** Relays the engine's fuel efficiency over a given distance.
- **Speed (km/h):** The vehicle's travel speed.
- **Carbon monoxide (CO):** CO concentration in the exhaust gases. It is measured as a percentage.
- **Hydrocarbons (HC):** Concentration of unburnt hydrocarbons in the exhaust. It is measured in parts per million (ppm).
- **Carbon dioxide (CO_2):** CO_2 concentration in the exhaust, indicative of combustion efficiency. It is measured as a percentage.
- **Oxygen (O_2):** Oxygen amount in the exhaust, offering insights into the combustion process. It is measured as a percentage.
- **Lambda:** The air–fuel equivalence ratio.
- **Air-Fuel Ratio (AFR):** The ratio of air to fuel in the combustion mixture.

3.1.2 Combinations

The variables are combined in three different subsets and evaluated across models.

1. **Engine-Only Models:** MAP, TPS, Force, Power, RPM, Consumption L/H, Consumption L/100KM, and Speed.
2. **Gas-Only Models:** CO, HC, CO2, O2, Lambda, and AFR.
3. **Variable Selection with Ridge Regression:** Ridge regression is used for variable selection in the third subset. The decision is expanded on below.

Using Ridge Regression for the third – variable-selection – subset is motivated by the engine fault dataset's structure: many predictors are likely relevant and strongly correlated due to physical couplings (e.g., intake/manifold dynamics driving gas-sensor responses). In that setting, Ridge is a better fit than LASSO because Ridge shrinks coefficients smoothly while keeping all features in the model, whereas LASSO encourages sparsity.

Ridge controls multicollinearity by distributing weight across correlated predictors rather than arbitrarily zeroing some of them out. This matters here because the goal is not to assume that only a small number of measurements carry signal; instead, the goal is to identify a reliable subset for downstream classification while avoiding the risk of discarding useful but redundant measurements. By design, LASSO can produce a sparse model with many exact zeros. That sparsity can translate into the removal of variables that are individually informative but jointly correlated—potentially hurting fault-discriminative information.

After Ridge shrinks coefficients, a retention threshold is applied to create the third subset while still benefiting from Ridge's stability. Variables are retained if their coefficient magnitude is above 0.1 times the maximum coefficient (i.e., threshold = max(abs(B)) $\times$ 0.1). This rule preserves comparatively influential sensors without forcing an overly aggressive "keep few" outcome. Compared to LASSO's implicit selection, this approach gives a more controlled mechanism: it uses Ridge to stabilize estimates under multicollinearity and then performs selection via an explicit magnitude cutoff. This is especially appropriate because the subsequent classification stage is expected to model nonlinear decision boundaries; therefore, it is beneficial that the variable subset supplied to classifiers is stable and not overly dependent on a sparsity assumption.

3.2 Models and Evaluation Methods

In the baseline, all features were retained and Decision Trees, Random Forest, Support Vector Classifier, KNN, and Neural Network performed comparably across Accuracy, Precision, Recall, and F1-Score [11]. The results suggest that there is potential for faults to be accurately classified with linear, non-linear, complex, and simple models. Considering past results, this analysis retains previous models and incorporates a range of new classifiers. Analysis is done in Python leveraging pandas, matplotlib, seaborn, sklearn, statsmodels, scipy, and numpy.

3.2.1 Logistic Regression (LR)

Logistic Regression predicts the probability that an observation is in a class. Observations should be independent, not have extreme outliers, and have a large sample size [8]. Additionally, there should be a linear relationship between the logit response and explanatory variables [10]. There should be no multicollinearity between predictors. In this case, the dependent variable has four possible outcomes, so multinomial logistic regression is used. Logistic regression is interpretable [1].

3.2.2 Decision Tree (DT)

A decision tree consists of a root node, branches, and internal and leaf nodes [5]. Branches are attribute values, internal nodes are attribute tests, and leaf nodes are final decision. The data is split on feature values, creating subsets where all items in a group belong to the same class. DTs are interpretable, non-linear, and robust to

outliers. However, they may overfit and be unstable.

3.2.3 Random Forest (RF)

Random Forest uses many decision trees to improve predictions. The results within trees are combined by voting to predict the class label. This helps improve accuracy, especially when there is a large enough dataset. RFs are robust to noise but computationally expensive, difficult to interpret, and may overfit [7].

3.2.4 Support Vector Machine Classifier (SVC)

Support Vector Machine finds the best hyperplane to separate classes. SVC typically performs well on high-dimensional data, regardless of outliers [6]. However, training can be slow and it can be difficult to interpret in high dimensions.

3.2.5 K-Nearest Neighbors (KNN)

KNN is a non-parametric method that classifies a point based on the majority class among its k nearest neighbors [3]. KNN is easy to implement but is slow with large data and can overfit when data contains noise.

3.2.6 Naive Bayes (NB)

Naive Bayes predicts classes from probability and assumes conditional independence of predictors given the class [4]. NB performs well on small training sets but assumes independent features and can generalize poorly.

3.2.7 Feed-Forward Neural Network (FFNN)

FFNNs have structured layers that data flows through [2]. Each layer has its weight adjusted to minimize prediction error. This implementation uses ReLU as the activation function due to its simplicity and efficiency. FFNNs can model complex patterns accurately but may overfit.

3.2.8 Quadratic Discriminant Analysis (QDA)

QDA relaxes LDA's assumption of a common covariance matrix, allowing each class to have its own covariance. This produces quadratic decision boundaries. QDA performs best when data is best separated with a curved boundary, the sample size is large, and classes have unique covariance structures. QDA is computationally heavy because classes have their own covariance matrix.

3.2.9 Linear Discriminant Analysis (LDA)

LDA assumes that each class follows a multivariate normal distribution with a shared covariance matrix. It finds a linear decision boundary by maximizing class separation. LDA performs best when classes can be separated with a linear boundary, the sample size is small, and the assumption of shared covariance is true. LDA is computationally light because of the single covariance matrix estimation.

3.2.10 Gradient Boosting (GB)

Gradient Boosting combines multiple weak decision trees into a single model. Base models are improved upon by training a new model on the errors. The combination allows GB to capture complex relationships, performing especially well on high-dimensional data. As DT and RF performed well on the baseline analysis, GB has the potential to correct errors and improve performance.

3.2.11 K-Fold Cross-Validation (CV)

K-fold cross-validation is done with the sklearn library and 5 folds to improve the robustness of model evaluation. Cross validation avoids overfitting and maximizes utilization of the dataset. The data is split into K folds and for each fold K-1 folds are used to train the model. The remaining fold serves as the test set. Accuracy,

Precision, Recall, and F-1 are calculated for each fold and averaged. Accuracy is the proportion of correct predictions made out of all predictions. Precision is the proportion of positive predictions that are correct. Recall is the proportion of actual positive cases out of true positive and false negatives. F-1 Score balances Precision and Recall.

3.2.12 Wilcoxon Signed Rank Test

This non-parametric statistical test compares two related groups by checking for significant difference between paired observations. In the case of model evaluation, pairs of models are evaluated on the same folds and metric. Wilcoxon is leveraged because the core assumption of a t-test – normality – is not met (Appendix A.1).

Name	Ridge Coefficients
CO2	-1.429820
O2	-1.018197
CO	-0.459436
Consumption L/100KM	-0.232141
MAP	-0.172101
Speed	-0.023660
HC	-0.002635
Force	-0.000158
RPM	0.000023
Power	0.011600
Lambda	0.048590
TPS	0.260664
Consumption L/H	0.395553
AFR	0.738967

Table 1: Variable Selection. Blue highlight indicates retained predictors.

4 Results

4.1 Variable Selection

Ridge Regression yielded several variables with coefficients close to 0, suggesting they have little effect on faults. Coefficients with magnitude greater than or equal to 0.1*max(abs(coefficients))=0.142 are kept. These include CO2, O2, CO, Consumption L/100KM, MAP, TPS, Consumption L/H, and AFR. Speed, HC, Force, RPM, Power, and Lambda are dropped before modeling in the variable selection model. Two of the dropped variables – HC and Lambda – are gas variables. The remaining four are engine variables.

4.2 Model Performance

The most notable models for performance measures are DT, RF, KNN, and GB. In the engine-only model, DT performs the best, GB performs second best, and RF performs worst across all accuracy measures (Table 2). As for the gas-only model, KNN performs best across all accuracy measures (Table 3). GB is second best for Precision, Recall and F1, while DT is third best for these three measures. DT is second best in Accuracy and RF is third best in Accuracy.

Finally, for the variable-selection model, GB performs best across all accuracy measures (Table 4) and models. For Precision, Recall, and F1, KNN performs second best and DT performs third best. DT is second best for Accuracy and KNN is third best in Accuracy.

The Wilcoxon Signed Rank Test was applied to the cross-validation folds for each variable grouping. LR, NB, FFNN, QDA, and LDA show strong evidence that different feature subsets produce different performance distributions at a p-level of 0.1; these show the most sensitivity to variable combinations (Figure 3). KNN and SVC show differences but not consistently across all comparisons. Lastly, DT, RF, and GB do not show significant differences between models – their performance is stable throughout variable subsets.

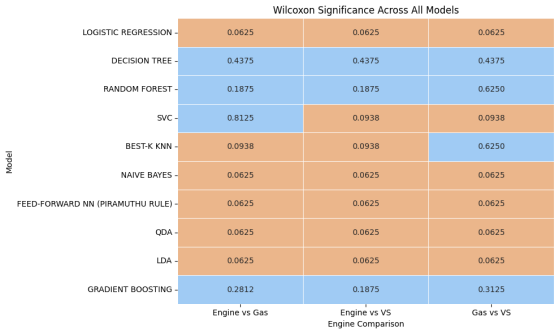


Figure 3: Wilcoxon Significance Test Results. P-values less than 0.1 in orange.

5 Analysis

5.1 Variable Selection

Variable selection via Ridge Regression provides evidence for why some models improve when going from the baseline – all features – to the variable-selected subset. Ridge shrinks coefficients of collinear predictors, so when multiple variables convey overlapping information, the model tends to suppress some of them toward zero. In the selected set, the retained features (AFR, TPS, CO2, O2, CO, MAP, Consumption L/100KM, Consumption L/H) have coefficient magnitudes above the retention threshold. Several high-VIF variables are removed, including Speed, Force, RPM, and Lambda (Appendix A.3). This pattern is consistent with multicollinearity: variables such as Lambda and AFR are strongly related, as seen in the correlation heatmap and discussed in Appendix A.2. Moreover, variables with coefficients near zero contribute little beyond what other correlated predictors already provide. Their removal reduces redundancy rather than discarding unique fault information. Removing redundant predictors can improve generalization – most notably for models that are sensitive to feature relationships.

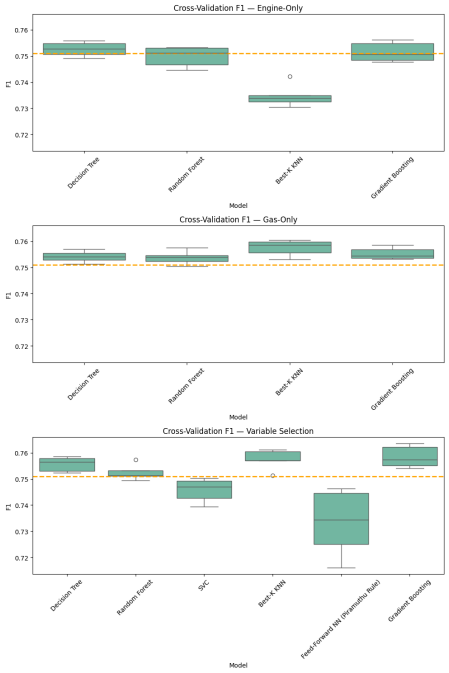


Figure 4: Cross-Validation F1 Performance across subsets and models, where orange line is the best performance in the baseline (Table 5). Since the baseline did not use CV, this comparison serves as evaluation method.

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.3871	0.4005	0.3971	0.3705
Decision Tree	0.7450	0.7527	0.7525	0.7526
Random Forest	0.7420	0.7496	0.7500	0.7498
SVC	0.6859	0.7039	0.6924	0.6899
Best-K KNN (4)	0.7393	0.7421	0.7436	0.7348
Naive Bayes	0.3463	0.2956	0.3286	0.2758
Feed-Forward NN	0.4966	0.4911	0.4899	0.4844
QDA	0.4520	0.5081	0.4832	0.4310
LDA	0.3960	0.4239	0.3951	0.3882
Gradient Boosting	0.7438	0.7514	0.7520	0.7516

Table 2: Engine-Only Model Performance – CV used. DT performs best across all accuracy measures.

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.5282	0.5568	0.5018	0.4898
Decision Tree	0.7458	0.7542	0.7541	0.7542
Random Forest	0.7455	0.7539	0.7539	0.7539
SVC	0.6914	0.6989	0.6977	0.6911
Best-K KNN (3)	0.7493	0.7575	0.7575	0.7575
Naive Bayes	0.4338	0.4533	0.4192	0.4181
Feed-Forward NN	0.5487	0.5637	0.5289	0.5109
QDA	0.5238	0.5208	0.5072	0.4820
LDA	0.5171	0.5404	0.4958	0.4942
Gradient Boosting	0.7470	0.7554	0.7554	0.7553

Table 3: Gas-Only Model Performance – CV used. KNN performs best across all accuracy measures.

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.5492	0.5545	0.5442	0.5465
Decision Tree	0.7474	0.7557	0.7557	0.7557
Random Forest	0.7437	0.7522	0.7522	0.7521
SVC	0.7402	0.7481	0.7493	0.7457
Best-K KNN (23)	0.7491	0.7574	0.7574	0.7574
Naive Bayes	0.4737	0.4798	0.4573	0.4472
Feed-Forward NN	0.6643	0.6649	0.6650	0.6646
QDA	0.6294	0.6528	0.6355	0.6262
LDA	0.5505	0.5584	0.5480	0.5520
Gradient Boosting	0.7501	0.7585	0.7586	0.7585

Table 4: Variable-Selection Model Performance – CV used. GB performs best across all accuracy measures.

5.2 Model Performance

LR, NB, FFNN, QDA, and LDA perform notably worse than DT, RF, SVC, KNN, and GB across variable subsets. In particular, their accuracy remains below .55 for the engine-only and gas-only model, suggesting these models struggle to learn fault-predictive decision boundaries under these variable sets. The varying performances are explained further, considering model assumptions, parameter tuning, and Wilcoxon Test results.

5.2.1 Model Assumption Violations

Logistic Regression assumes a linear relationship between features and the log-odds of the target and is sensitive to multicollinearity. In this setting, variables are inherently related, which is reflected in the correlation heatmap (Appendix A.2). For example, TPS is highly correlated with Force and MAP. TPS and MAP are physically related because throttle opening directly changes the amount of air that flows into the intake manifold, which affects Force. These dependencies likely reduce the stability of the learned coefficients, contributing to poor generalization. Notably, LR performs better in the gas-only and variable-selection settings than in the engine-only setting, suggesting that the engine feature subset may contain stronger or more pervasive redundancy. Additionally, the variable selection procedure likely reduced some multicollinearity, which may have partially improved performance. Overall, LR is among the weakest performers, consistent with the baseline findings reported in the original paper (Table 5).

QDA, LDA, and Naive Bayes rely on distributional assumptions that appear violated by the observed feature distributions. Skewness and lack of symmetry are depicted in the histograms and box plots (Appendix A.1); normality assumptions do not hold. The violation helps explain why these models do not capture the underlying structure effectively in the full feature set. Interestingly, QDA, LDA, and NB show slightly improved performance in the gas-only and variable-selection models, and Naive Bayes even exceeds the original model results in the gas-only and variable-selection settings (Table 5). This suggests that restricting to gas variables or removing unhelpful redundancy makes the remaining feature distributions more compatible with their assumptions.

5.2.2 Parameter Tuning Challenges

SVC and the neural network perform worse than the baselines across many settings. The results appear sensitive to parameter choices. For the neural network, computational constraints limited tuning; the model therefore used a fixed hidden-layer structure selected via the Piramuthu rule, with common defaults (ReLU activation, Adam optimizer, and learning rate of 0.01). Insufficient tuning can prevent the network from fitting the nonlinear fault boundaries accurately. Similarly, SVC is highly sensitive to kernel-related hyperparameters and to feature scaling. With limited tuning, the learned decision boundary does not align with the true fault boundaries. Additionally, SVC performs well in high-dimensions. The engine-only and gas-only models include fewer predictors than the baseline and variable-selected model, which may contribute to the mixed results. Overall, the pattern indicates that both SVC and the FFNN require more careful hyperparameter optimization to reach competitive performance under the current experimental setup.

5.2.3 Competitive Models

For the remaining models (DT, RF, KNN, and GB), results are stronger and more consistent, particularly for Precision, Recall, and F1. In the engine-only subset, DT provides the best performance among the evaluated methods, achieving higher Precision, Recall, and F1 than the DT baseline (Table 5). RF improves on the RF baseline across all model configurations, with consistent gains in Precision, Recall, and F1. SVC is weaker across most metrics compared to the top-performing models, except for Recall and F1

Model	Accuracy	Precision	Recall	F1
LR	0.576	0.574	0.576	0.574
DT	0.750	0.750	0.750	0.750
RF	0.748	0.748	0.748	0.748
SVC	0.747	0.768	0.747	0.715
KNN	0.751	0.751	0.751	0.751
NB	0.394	0.370	0.394	0.353
Neural Net.	0.749	0.748	0.749	0.748

Table 5: **Baseline** performances from original paper [11]. The authors computed on a test set from 80/20 split, whereas **this analysis uses CV.**

in the variable-selection setting. KNN performs better than the baseline mainly in the gas-only and variable-selection models, where its Precision, Recall, and F1 improve. Finally, GB outperforms the baseline across Precision, Recall, and F1 for all evaluated settings, indicating that boosted decision trees capture nonlinear relationships that are important for fault classification more effectively than the alternative learners and assumption-driven models.

5.2.4 Wilcoxon Test with F1-Score

Because the objective is fault classification, Precision, Recall, and F1 are the most important metrics. Accuracy can appear acceptable even when one class is systematically misclassified. Recall ensures critical faults are not missed – fewer false negatives – while Precision avoids excessive false alarms – fewer false positives). F1 balances both, providing the most informative single summary of performance for this fault-detection task. Because of this, F1 is used as the metric for the Wilcoxon Test.

The results shown in the Wilcoxon Test are consistent with the aforementioned results (Figure 3). Models that rely on underlying assumptions, distance-based metrics, or parameter tuning show significant differences across variable subsets (LR, NB, FFNN, QDA, LDA), while DT, RF, and GB do not show differences across any pair.

6 Conclusion

6.1 Summary

This project strengthens earlier engine-fault classification work by addressing a key methodological gap: the baseline analysis evaluates multiple classifiers but does not perform variable selection despite clear multicollinearity and physically grounded relationships among measurements. In contrast, this paper compares three distinct variable subsets—engine measurements, gas-sensor measurements, and a Ridge Regression–selected set—and evaluates an expanded set of model families. This design clarifies whether the signal used for fault prediction is inherently tied to specific sensor groups or whether combining variables is redundant.

Equally important, this study improves the evaluation methodology. The baseline results rely solely on accuracy, which does not characterize performance stability across different train/test splits. Here, results are assessed using cross-validation, providing a more reliable estimate of performance under varying data partitions. Furthermore, significance is tested using a Wilcoxon procedure grounded in the F1-score; this better reflects fault-detection priorities by jointly accounting for precision and recall. Under this cross-validated evaluation, the gas-only and variable-selection setups show consistent, measurable improvements: multiple models outperform the baseline F1 across several folds, not just

in a single split or under assumptions that may not hold.

Model behavior across variable subsets also yields practical guidance. Several models—including Logistic Regression, Naive Bayes, the feed-forward neural network, QDA, and LDA—tend to be weaker or more sensitive to the chosen variable representation, aligning with issues such as violated distributional assumptions and limited ability to capture nonlinear fault boundaries. For researchers specifically interested in stability across variable sets, Decision Trees, Random Forests, and Gradient Boosting are recommended. These models are competitive in performance across all evaluated subsets and did not show significant differences across models in the Wilcoxon Test—suggesting stronger robustness when changing the sensor representation. This stability makes them particularly attractive for building and comparing vehicle-health monitoring pipelines where the availability or relevance of sensors may vary by application or platform.

Finally, the findings translate into actionable sensor prioritization recommendations. For car manufacturers, this study supports focusing development on and ensuring onboard diagnostic access to the following measurements for fault-detection flagging: CO2, O2, CO, Consumption L/H, Consumption L/100KM, MAP, and AFR. For car buyers, these results motivate checking whether these variables are monitored and available to the user-side through dashboards, diagnostic access, or compatible reporting tools; availability of these channels can directly affect the effectiveness and usefulness of fault detection in real-world ownership.

Taken together, the results provide actionable guidance for both diagnostic system design and further research. First, gas-sensor measurements and selected variable subsets contain fault-predictive signal, but the strongest overall performance comes from nonlinear learners that can model complex interactions without requiring strict statistical assumptions. Second, for onboard diagnostic prioritization—where operational reliability matter—Gradient Boosting and Random Forests are strong candidates for accurate and robust fault detection using the tested sensor feature sets. More broadly, the study supports the value of combining principled variable selection with model evaluation backed by cross-validation and significance testing, improving confidence in which modeling approaches and measurement groups are most appropriate for real-world engine-health monitoring.

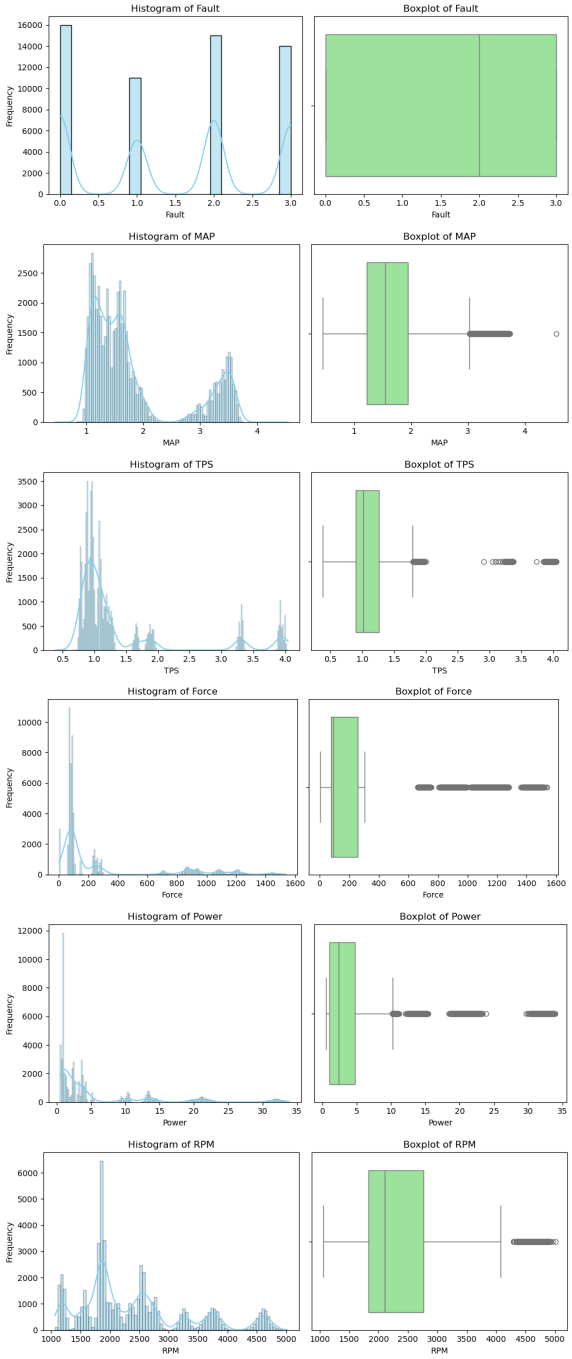
6.2 Lessons Learned

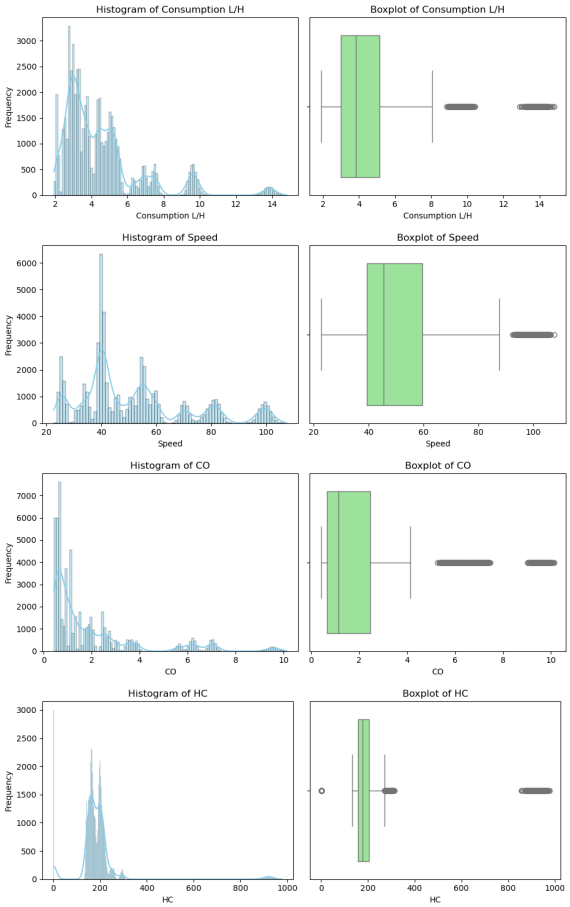
This course helped me develop a rigorous process for data analysis. I enjoyed getting feedback from peers and TAs because many people pointed out problems or improvements I didn't originally see. I incorporated feedback from past homework into new ones and the project, and I will carry these lessons moving forward in my career.

One note for improvement is that some of the reports I have read are very low effort and wouldn't be tolerated (or even submitted) in my undergraduate program. While I can't see the final grades on these, I hope the grading is fair and reflects both depth and correctness. I wasn't able to find example reports on the Canvas page, but I think including them would establish a higher baseline expectation for work submitted – we all learn more when the whole community is giving their best effort.

A Figures

A.1 Histograms and Boxplots





A.2 Correlation Heatmap

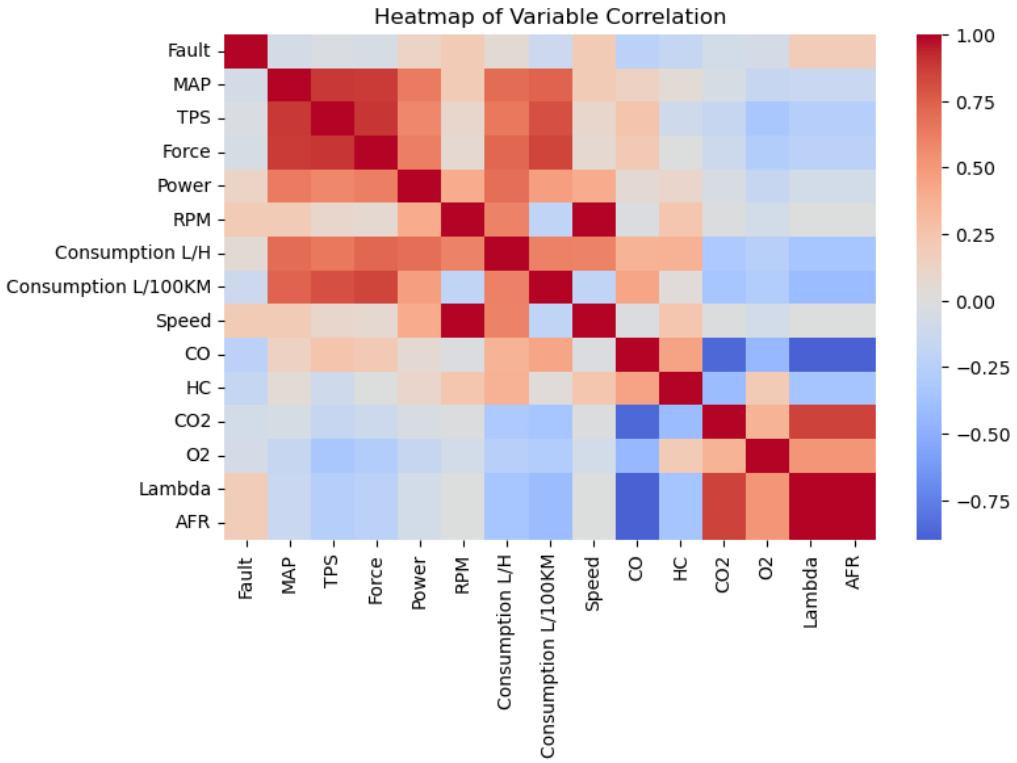


Figure 5: Correlations of features. Potential multicollinear relationships include CO & AFR, Lambda, & HC, MAP & TPS & Force, Speed & RPM.

Discussion:

The multicollinearity within engine-only and gas-only variables is evident from the heatmap. The physical relations between variables provide an explanation. For example, TPS, MAP, Power, Force, and Consumption l/H exhibit high correlations with each other. Note TPS and MAP: TPS (throttle opening) determines how much air can enter; that airflow raises MAP (manifold absolute pressure). That overlap shows up as high correlation. Further, higher MAP indicates higher intake which supports higher Force generation; Power rises with Force; higher Power and Force increases fuel burned per hour, increasing Fuel Consumption L/h.

Exhaust variables also have explainable relationships. Lambda and AFR have a direct relationship. A lean mixture indicates higher AFR and Lambda less than 1, while a rich mixture indicates lower AFR and Lambda greater than 1. If the mixture is lean, more Oxygen is in the exhaust.

A.3 VIF Table

Table 6: VIF and Multicollinearity Assessment

Variable	**VIF**	**Multicollinearity**
AFR	52064.931965	high
Lambda	52057.734318	high
RPM	168.748555	high
Speed	164.461881	high
Consumption L/H	36.357656	high
Consumption L/100KM	25.763986	high
Force	11.245738	high
MAP	8.618050	high
TPS	8.571702	high
CO	8.489502	high
CO_2	5.035379	high
HC	4.412623	moderate
Power	2.280765	moderate
O_2	2.161125	moderate

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